



Disease note

Four new fungal pathogens causing avocado dieback in Brazil

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ABSTRACT

Dieback is a disease with a complex etiology that results in the death of branches or entire plants in avocado producing regions. Losses associated with this disease have already been reported in the main producing regions of Brazil. The etiological agents of this disease are different and an association between fungi is often observed. This study aimed to identify fungi associated with avocado dieback in Brazil. Fungal isolates were collected from symptomatic avocado stems from orchards in São Paulo state. Identification was based on phenotypic characteristics and multilocus phylogenetic analysis of the translation elongation factor 1-alpha, RNA polymerase II, β -tubulin, actin, the internal transcribed spacer (ITS) region of rDNA, and the large subunit ribosomal RNA regions, according to the previously identified genus. The pathogenicity test confirmed that all species were pathogenic to avocado seedlings and fruits. This is the first report of *Neopestalotiopsis arecaeum*, *Neocosmospora bostryoides*, *Nectria pseudotrichia* and *Cytospora viridistroma* causing avocado dieback in Brazil and worldwide. Identifying emerging pathogens that cause losses is crucial for avocado-producing countries with different climatic conditions, as it helps understand the geographic distribution of these phytopathogens and the conditions that favor their development.

1. Introduction

The avocado is an important fruit tree distributed mainly in the Midwest, Southeast, and South of Brazil. Most fruit production is commercialized for domestic consumption, while only a small portion is exported (Instituto Brasileiro de Geografia e Estatística, 2024; Sommaruga and Eldridge, 2021). Its nutritional value and unique characteristics make it highly attractive to the pharmaceutical industry, positioning it as the fastest-growing commodity among major tropical fruits (FAO, 2021; Flores et al., 2019).

Avocado trees can be impacted by various diseases both in the field and post-harvest, resulting in significant losses in production and commercialization (Piccinin and Pascholati, 2018; Fischer and Firmino, 2023). Dieback is a disease with a complex etiology that causes leaf yellowing, wilting, and necrosis, followed by the gradual death of branches and stems. It begins with the drying of the tips and can lead to the death of entire branches and trees (Hultberg et al., 2020; Valor et al., 2020). It can be caused by both biotic and abiotic factors, but primarily by the combination of both (Camarero et al., 2015). The colonization of the xylem by fungi limits the accumulation of nutrient reserves, which

can hinder fruit development or even result in the death of the plant (Auger et al., 2013). In addition, infected fruits may develop mycelial growth or necrotic lesions, which reduces their shelf life or makes them unsuitable for commercialization (Valencia et al., 2019).

Taxonomic studies using molecular tools in several countries revealed different etiological agents (Fiorenza et al., 2023; Guarnaccia et al., 2018; Hernández et al., 2023; Lin et al., 2018; Piccinin and Pascholati, 2018; Ploetz et al., 2017; Ramírez-Gil and Peterson, 2019; Trakunyingcharoen et al., 2015; Valencia et al., 2022). Despite the high diversity of fungal species, traditionally the diagnosis of diseases is based only on the symptoms observed in the field. In Brazil, *Colletotrichum*, *Lasiodiplodia*, *Nectria*, *Neofusicoccum*, *Phytophthora*, *Rosellinia*, and *Verticillium* have been reported to cause dieback in avocado (Piccinin and Pascholati, 2018; Fischer and Firmino, 2023). However, only *Colletotrichum* has been identified through phylogenetic analyses (Oliveira et al., 2024; Soares et al., 2021., 2021). Accurate identification of phytopathogens enables effective disease management. The monitoring of avocado diseases revealed plants exhibiting symptoms of dieback. Therefore, this study aimed to identify fungal species associated with avocado dieback in Brazil and to assess their pathogenicity in

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avocado seedlings and fruits.

2. Materials and methods

2.1. Sampling and isolation

The collections were carried out on branches of seven avocado trees in producing areas of Mogi Mirim - SP in 2020 (22° 25' 55"S, 46° 57' 28"O). Briefly, pieces of approximately 1 cm of symptomatic tissue were superficially sterilized with 70% alcohol for 1 min and then in 1% hypochlorite for 2 min, followed by consecutive washes in distilled water three times. The symptomatic tissues were then transferred to Petri dishes containing Potato-Dextrose-Agar (PDA) medium. The plates were incubated at 25 ± 2 °C in the dark. Emerging hyphal tips from the infected tissues were transferred to new PDA plates, and the pure cultures of seven fungal isolates were stored in the Coleção de Culturas da Universidade de Brasília (CCUB).

2.2. Cultural and morphological characterization

Morphological analyses were performed using a single isolate from each identified species. The fungal colonies were grown on PDA for color evaluation. Agar-agar (AA), synthetic nutrient-poor agar (SNA; 1000 mL H₂O, KH₂PO₄ 1 g, KNO₃ 1 g, MgSO₄·7H₂O 0.5 g, KCl 0.5 g, glucose 0.2 g, sucrose 0.2 g, and agar 20 g), and avocado branches were added when necessary to induce sporulation. The colonies were incubated at 25 °C with 12 h of light and 12 h of darkness. Fungal structures were mounted in 10% lactic acid for microscopic observation. Images were obtained with a Leica DFC 490 digital camera attached to a Leica DM 2500 light microscope (Leica Microsystems, Nussloch, Germany). Image capture and measurement were performed using Leica Qwin-Plus.

2.3. DNA extraction and sequencing

DNA extractions from all seven fungal isolates were performed using the Wizard Genomic DNA Purification Kit (Promega Corporation, WI, U. S.A) according to [Pinho et al. \(2013\)](#). Firstly, the region of the translational elongation factor (TEF1-α) was amplified for all isolates for fungal genus identification. Subsequently, for the multilocus analysis, the regions RNA Polymerase II (RPB2), β-tubulin (BTUB), actin (ACT), internal transcribed spacer of rDNA (ITS), and large subunit ribosomal ribonucleic acid (LSU) were sequenced according to the genus previously identified. The PCR conditions for each genomic region are detailed in [Table 1](#). PCR products were purified and bidirectionally Sanger-sequenced.

2.4. Phylogenetic analysis

Previous study sequences were obtained from GenBank

([Supplementary Table 1](#)) and aligned using multiple sequence alignments performed with MAFFT version 7 ([Katoh and Standley, 2013](#)). Alignments were concatenated in Sequence Matrix 1.8 ([Vaidya et al., 2011](#)). Phylogenetic relationships were inferred using maximum likelihood (ML) in RaxML-HPC2 ([Stamatakis, 2014](#)). Bayesian inference (BI) was conducted using the best-fit models of nucleotide substitution selected according to the corrected Akaike information criterion (AIC). The matrices were submitted by the Monte Carlo method, using Markov Chain (MCMC) carried out in MrBayers 3.1.2 ([Ronquist and Huelsenbeck, 2003](#)) within the CIPRES platform ([Miller et al., 2010](#)) under analysis of 10 million generations with sampling every 1,000, with subsequent discarding of 20% of the first trees under analysis. The trees of *Neocosmospora*, *Neopestalotiopsis*, *Cytospora*, and *Nectria* were rooted with *Fusarium cicatricum*, *Pestalotiopsis theae*, *Cytospora rostrata*, and *Cosmospora coccinea*, respectively.

2.5. Pathogenicity test

The test was conducted on 6-month-old Margarida avocado seedlings and green Hass fruits. The fruits were washed with 70% alcohol for 1 min and 2% hypochlorite for 2 min, then rinsed with water three times. Six-millimeter wounds were made in the pulp of the fruits and in the stem of the seedlings for inoculation. The isolates were grown on PDA for 7 days, and 5 mm mycelium discs taken from the colony's edge were placed in the wounds and secured with damp cotton and Parafilm®. The negative control used only PDA. The fruits were kept in boxes in a humid chamber for two days. The plants were maintained in a greenhouse at a temperature of 30/18 °C day/night with a photoperiod of 16/8 h day/night. Pathogenicity was checked 5 and 40 days after inoculation for the fruits and seedlings, respectively. Symptoms of wilting or darkening of the vascular system in the seedlings and darkening of the fruit were assessed. Subsequently, the fungi were reisolated. The tests were independently repeated twice.

3. Results

3.1. Sampling and fungal isolation

Two plantations were evaluated in the main producing region of São Paulo. Symptoms observed included wilting, severe leaf drop, and dieback in avocado trees, as well as necrotic internal tissues. The collected samples resulted in seven isolates with typical characteristics of the Nectriaceae, Pestalotiopsidaceae, and Cytosporaceae families ([Table 2](#)). All isolates were selected for phylogenetic analyses and pathogenicity tests.

3.2. Phylogenetic analysis and morphological characterization

Based on the first approximation of taxonomic identity by BLASTn,

Table 1
List of primers that were used in this study.

Locus	Primer name	sequence (5'–3')	Orientation	(Annealing/Extension)	Reference
Elongation factor 1-α (EF1-α)	EF1F1	TGCGGTGGTATCGACAAGCGT	Forward	56°-45''/72°-45''	Jacobs et al. (2004)
	EF2R1	AGCATGTTGTCGCCGTGAAG	Reverse	56°-45''/72°-45''	Jacobs et al. (2004)
RNA polymerase II second largest subunit (RPB2)	5F2	GGGGWGAYCAGAAGAAGG	Forward	54°-45''/72°-45''	Sung et al. (2007)
	7Cr	CCCATRGCCTTGYTTRCCCAT	Reverse	54°-45''/72°-45''	Liu et al. (1999)
β-tubulin (BTUB)	T1	AACATGCGTGAGATTGTAAGT	Forward	55°-45''/72°-45''	O'Donnell and Cigelnik (1997)
	Bt2-b	ACCCTCAGTGTAAGTACCCTTGGC	Reverse	55°-45''/72°-45''	Glass and Donaldson (1995)
28S nrRNA (LSU) e Internal transcribed spacer (ITS)	V9G	TTACGTCCCTGCCCTTTGTA	Forward	53°-45''/72°-45''	Vilgalys and Hester (1990)
	LR5	TCCTGAGGGAAACTTCG	Reverse	53°-45''/72°-45''	De Hoog and Gerrits Van Den Ende (1998)
Actin (ACT)	ACT-512F	ATGTGCAAGGCCGGTTTCGC	Forward	58°-45''/72°-25''	Carbone and Kohn (1999)
	ACT-783R	TACGAGTCCTCTGCGCCCAT	Reverse	58°-45''/72°-25''	Carbone and Kohn (1999)

Table 2

List of families of Brazilian avocado isolates included in this study.

Family	Isolate code ^a	Local
Nectriaceae	CCUB 3537	Mogi Mirim (SP) 22° 25' 55" S, 46° 57' 30" W
	CCUB 3542	Mogi Mirim (SP) 22° 25' 55" S, 46° 57' 30" W
	CCUB 3530	Mogi Mirim (SP) 22° 25' 55" S, 46° 57' 30" W
Pestalotiopsidaceae	CCUB 3534	Mogi Mirim (SP) 22° 25' 55" S, 46° 57' 30" W
	CCUB 3545	Mogi Mirim (SP) 22° 25' 55" S, 46° 57' 30" W
Cytosporaceae	CCUB 3528	Mogi Mirim (SP) 22° 25' 55" S, 46° 57' 30" W
	CCUB 3490	Santo Antonio de Posse (SP) 22° 36' 24" S, 46° 55' 9" W

^a Isolate code from the Coleção de Cultura da Universidade de Brasília (CCUB) at the Universidade de Brasília (UnB), Brazil.

using the TEF1- α sequence, the isolates were assigned to four genera: *Neocosmospora* (CCUB 3537 and 3542), *Neopestalotiopsis* (CCUB 3534 and 3545), *Nectria* (CCUB 3530) and *Cytospora* (CCUB 3490 and 3528).

The TEF and RPB2 genes were utilized for the identification of *Neocosmospora* species, resulting in matrices containing 43 taxa. The concatenated alignment included 1381 sites, of which 1284 were conserved and 97 were informative for parsimony. The results confirmed that the isolates grouped with *N. bostrycoides* were well-supported in both ML and BI multilocus analyses (Fig. 1). The pure cultures of isolates CCUB 3542 and 3537 showed typical morphology of *Neocosmospora* sp. with yellow coloration and white aerial mycelium in both cultures. In PDA, microconidia formation was seen from the 5th day of cultivation with a size of $3.5\text{--}7.0 \times 2.0\text{--}4.0 \mu\text{m}$ ($5.28 \pm 0.91 \times 2.91 \pm 0.4 \mu\text{m}$, $n = 30$). The production of macroconidia was only seen in synthetic nutrient-poor agar after 15 days in the dark, with a size of $25.5\text{--}34.5 \times 2.5\text{--}4.0 \mu\text{m}$ ($31.01 \pm 2.06 \times 3.09 \pm 0.26 \mu\text{m}$, $n = 30$) with five to six septa (Fig. 5A–C). Chlamydospores were not observed.

The combined matrix of the genes TEF1- α , BTUB, and ITS was used

for the genus *Neopestalotiopsis*, consisting of 31 taxa, totaling 2221 characters, of which 77 were informative for parsimony and 2144 were conserved. After the analysis, it was confirmed that the isolates CCUB 3534 and 3545 grouped with *N. araccacearum* (Fig. 2). Isolates CCUB 3534 and 3545 showed colonies with radial growth, with a cotton white color observed on the upper side and light yellow on the reverse side, confirming the morphology of *Neopestalotiopsis*. Cultures were covered with dark cells after 14 days of incubation. Conidia were fusiform to ellipsoid, with CCUB 3534 exhibiting a length and width of $19.5\text{--}28.5 \times 5.5\text{--}8.0$ ($24.20 \pm 2.26 \times 6.75 \pm 0.26 \mu\text{m}$, $n = 30$), with 5 versicolor cells, hyaline apical and basal cells and three brown median cells. They have between two and four apical and one basal appendage (Fig. 5D–F).

The ML analyses combined with BI of the individual TEF1- α partitions for the *Nectria* isolate included 31 ingroup taxa. The alignment resulted in 798 bp, with 212 parsimony-informative sites. The posterior probabilities of BI were plotted on the ML tree, indicating that CCUB 3530 belongs to *N. pseudotrachia* with high branch support (Fig. 3). The *Nectria* isolate grew rapidly on PDA, and after 11 days, the entire plate was completely covered with a white and brown colony against a red-dish background. No reproductive structure was observed (Fig. 5L).

The genes ITS, ACT, LSU, RPB2, TEF1- α , and BTUB were used for the phylogenetic analysis of the *Cytospora* isolates, including 40 taxa, resulting in a concatenated alignment of 3741 characters, of which 701 were conserved and 2688 were informative for parsimony. In our analysis of individual trees using ML and BI (Supplementary material), the isolates CCUB 3490 and 3528 were grouped into a single clade containing isolates of *C. viridostroma* (Fig. 4). The Isolates CCUB 3490 and 3528 showed characteristic morphology of *Cytospora* with growth on PDA in 7–10 days with white, flat color, uniform texture, and regular growth. Sexual reproduction structures were not visualized. The asexual structure was visualized in the avocado stem in PDA after 40 days of cultivation under fluorescent light. Pycnidia were dark brown with a

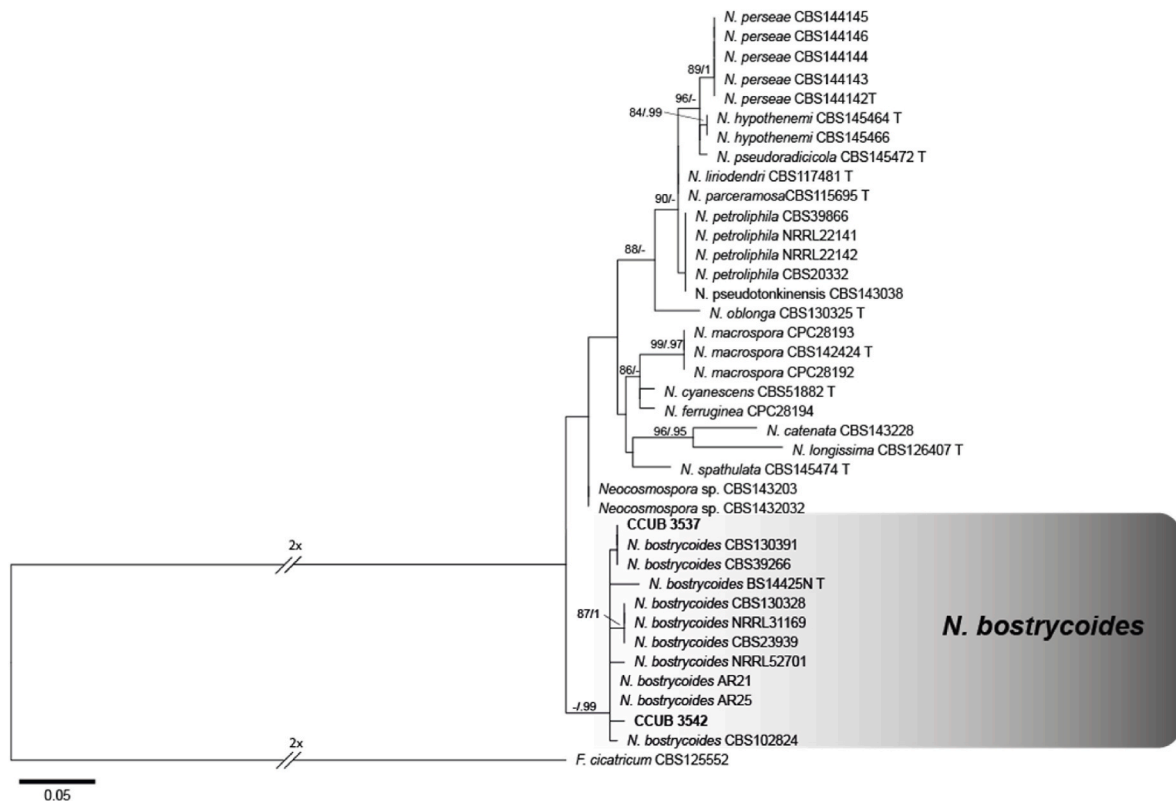


Fig. 1. The Bayesian inference tree of the *Neocosmospora* from a concatenated alignment of TEF1- α and RPB2. Significant SH-aLRT bootstrap supports (≥ 80) and posterior Bayesian probability (≥ 95) are shown above the nodes. The tree was rooted with *F. cicatricum*. Ex-type isolates are indicated with “T” at the end of taxon labels. Isolates from the present study are highlighted in bold font. The scale bar indicates the average number of substitutions per site.

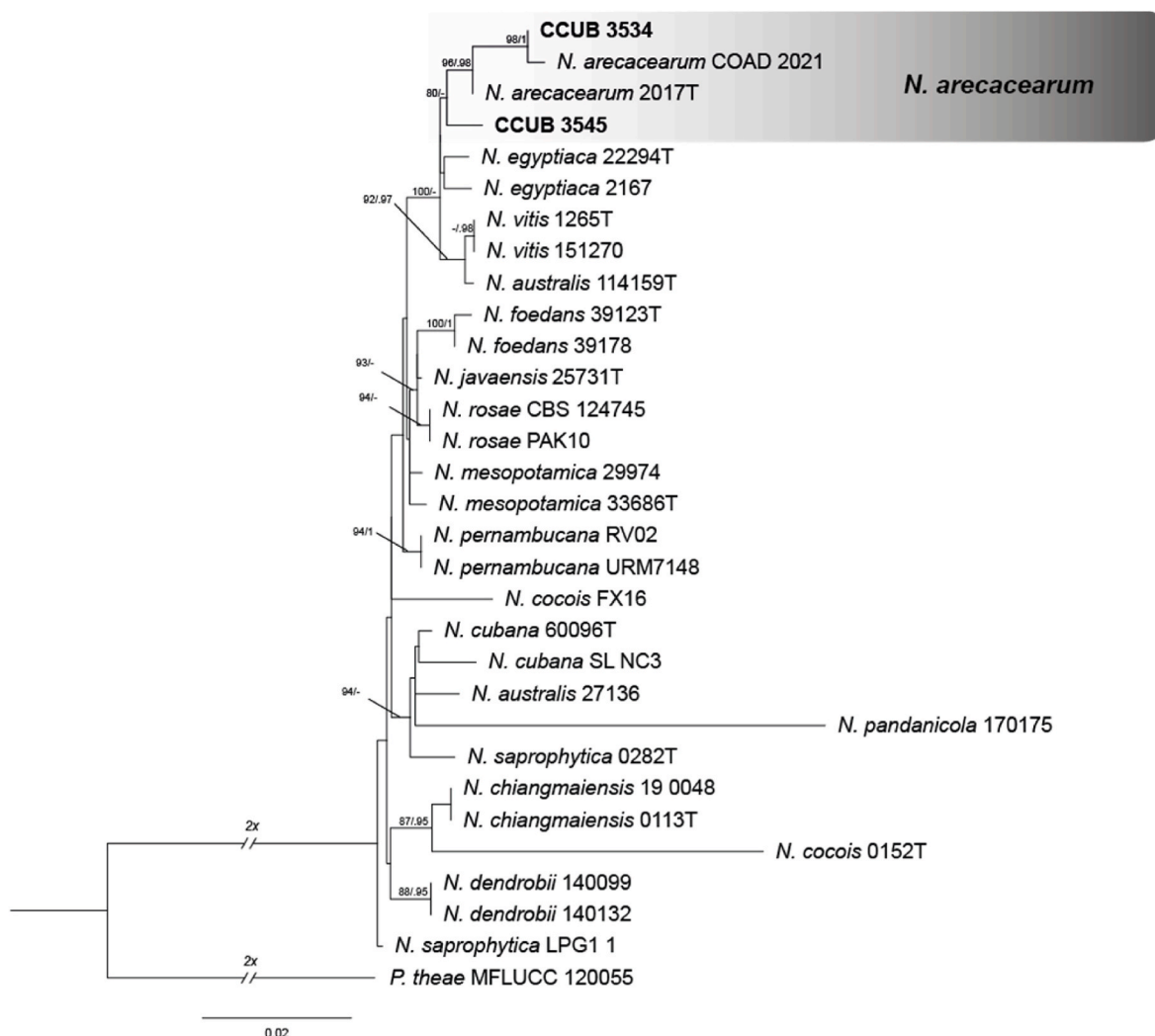


Fig. 2. The Bayesian inference tree of the *Neopestalotiopsis* from a concatenated alignment of TEF1- α , BTUB and ITS. Significant SH-aLRT bootstrap supports (≥ 80) and posterior Bayesian probability (≥ 95) are shown above the nodes. The tree was rooted with *Pestalotiopsis theae*. Ex-type isolates are indicated with “T” at the end of taxon labels. Isolates from the present study are highlighted in bold font. The scale bar indicates the average number of substitutions per site.

size of $180.0\text{--}310.5 \times 126.5\text{--}182.0 \mu\text{m}$ ($246.80 \pm 57.69 \times 159.61 \pm 26.29 \mu\text{m}$, $n = 20$). Unbranched hyaline conidiophores, aseptate hyaline conidia with CCUB 3490 sizes of $2.5\text{--}4.0 \times 1\text{--}1.5$ ($3.48 \pm 0.27 \times 1.40 \pm 0.17$, $n = 50$) (Fig. 5G–K).

3.3. Pathogenicity test

All the isolates were pathogenic to the avocado fruit and seedlings. The symptoms observed on the surface of the fruits were dark brown necrotic lesions around the inoculation sites. In the seedlings, there was darkening of the vascular tissue, indicating pathogen colonization. In the control, seedlings and fruits remained asymptomatic (Fig. 6).

4. Discussion

In Brazil, dieback is commonly observed in commercial avocado plantations. However, causal agents have not yet been accurately identified through phylogenetic analysis. Considering that the symptoms result from the combined effects of plant stress and the attack of fungi, it is assumed that different genera and species can cause dieback (Hrycan et al., 2020; Jurskis, 2005). In this study, isolates of *Neocosmospora*, *Neopestalotiopsis*, *Cytospora* and *Nectria* were recovered from *P. americana* showing symptoms of dieback between 2020 and 2022 in

Brazil. Pathogenicity tests, combined with morphological characteristics and phylogenetic analyses, confirm that *Cytospora viridistroma*, *Nectria pseudotrichia*, *Neocosmospora bostrycoides*, and *Neopestalotiopsis areacearum* cause dieback and fruit rot in avocados. All species are reported for the first time on avocado in Brazil.

In this study, *Neocosmospora bostrycoides* cause disease in the stem and fruit of avocado. In Brazil, this fungus was only reported on yellow passion fruit plants (Ninos et al., 2021) but has been reported in other hosts including coffee and *Ficus carica* in Puerto Rico and Italy, respectively (Gusella et al., 2024; Serrato-Diaz et al., 2024). Although *Neocosmospora euwallaceae* have been reported on avocados in South Africa and *Neocosmospora perseae* in Italy and Greece (Guarnaccia et al., 2022, 2018; van den Berg et al., 2019), this is the first report of *Neocosmospora bostrycoides* causing avocado dieback worldwide.

Neopestalotiopsis siciliana and *N. rosae* have previously been reported to cause dieback on avocado in Italy and Turkey, respectively (Çalış et al., 2024; Fiorenza et al., 2022). Although *Neopestalotiopsis areacearum* has been previously reported in Brazil on *Caryota mitis*, *Dypsis lutescens*, *D. madagascariensis*, and *Ptychosperma elegans* (Guterres et al., 2023), this is the first global report of this species causing dieback in avocado.

Nectria pseudotrichia has been reported to cause stem-end rot in avocado fruits in Kenya (Wanjiku et al., 2020). In Brazil, this species has

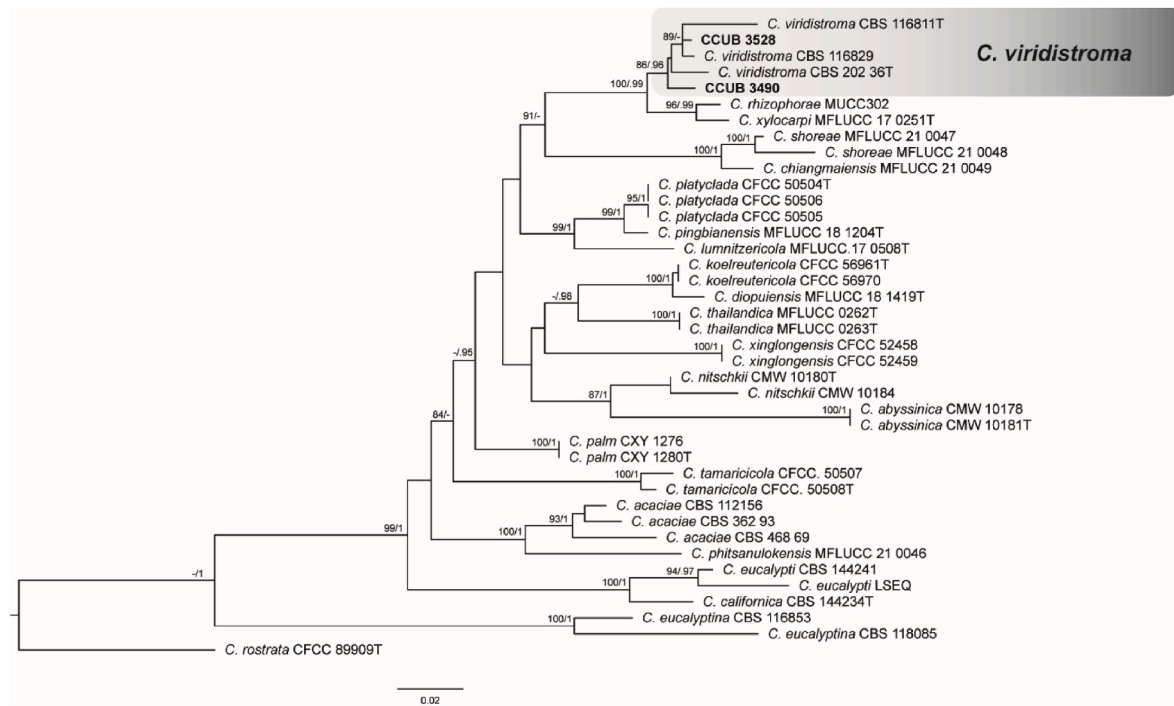


Fig. 3. The Bayesian inference tree of the *Nectria* from a concatenated alignment of ITS and TEF1- α . Significant SH-aLRT bootstrap supports (≥ 80) and posterior Bayesian probability (≥ 95) are shown above the nodes. The tree was rooted with *Cosmospora coccinea*. Ex-type isolates are indicated with “T” at the end of taxon labels. Isolates from the present study are highlighted in bold font. The scale bar indicates the average number of substitutions per site.

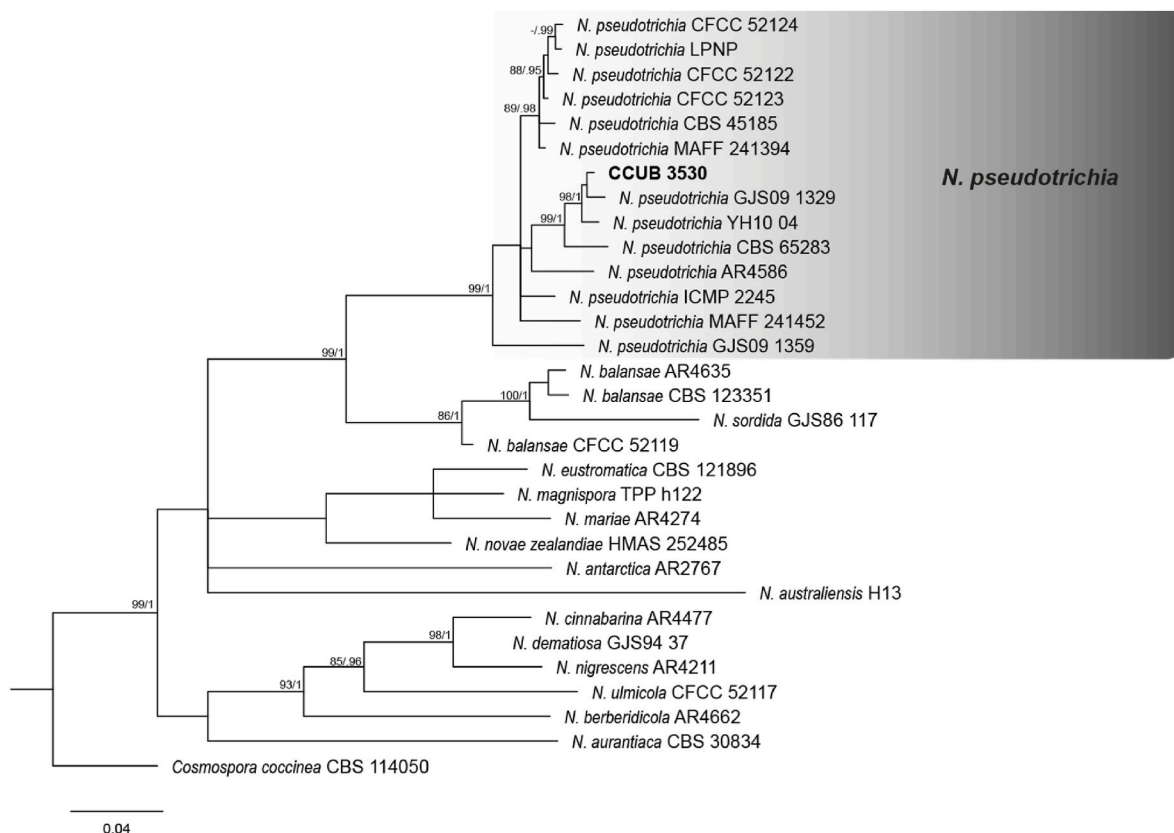


Fig. 4. The Bayesian inference tree of the *Cytospora* from a concatenated alignment of ITS, ACT, LSU, RPB2, TEF1- α , and BTUB. Significant SH-aLRT bootstrap supports (≥ 80) and posterior Bayesian probability (≥ 95) are shown above the nodes. The tree was rooted with *C. rostrata*. Ex-type isolates are indicated with “T” at the end of taxon labels. Isolates from the present study are highlighted in bold font. The scale bar indicates the average number of substitutions per site.

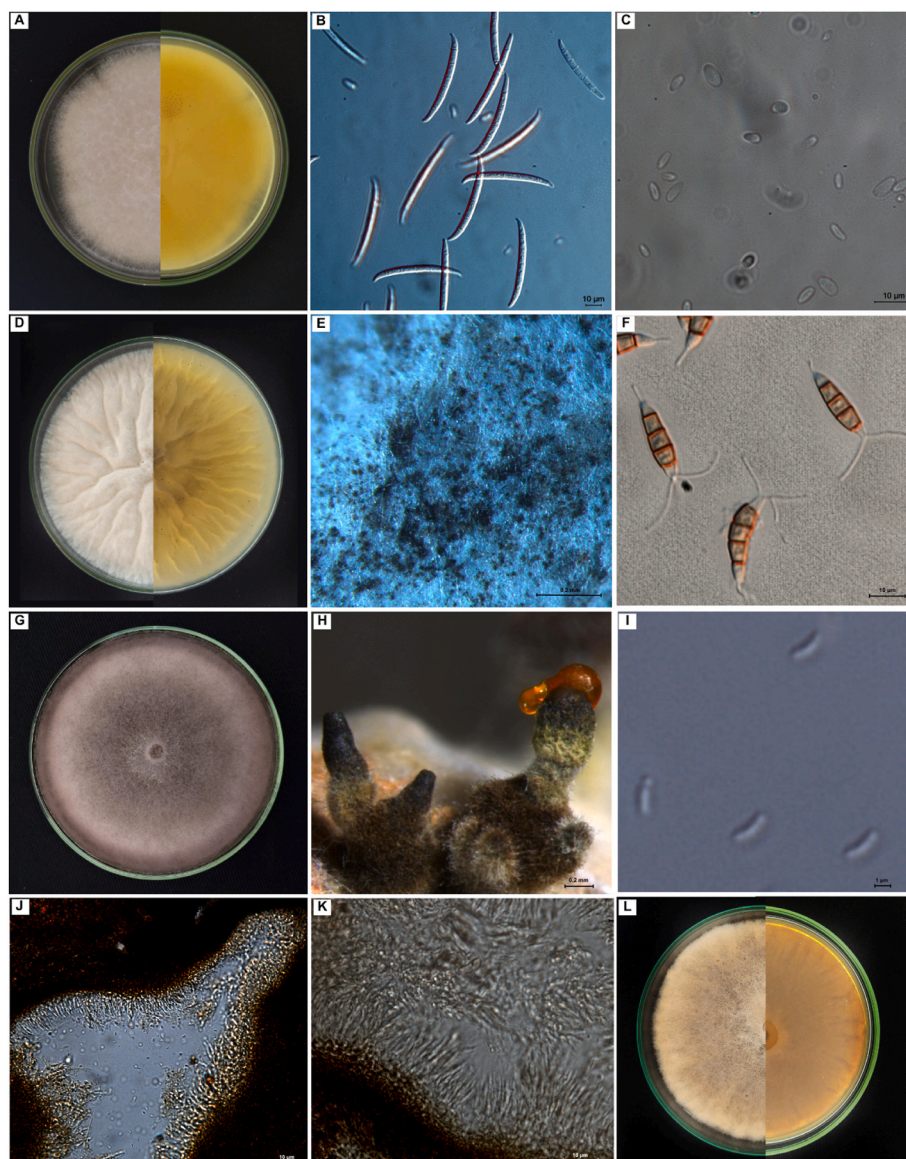


Fig. 5. Morphological characteristics of *Neocosmospora bostrycoides* (A–C), *Neopestalotiopsis aracacearum* (D–F), *Cytospora viridistroma*. (G–K) and *Nectria pseudotrachia* (L). Front and back from culture of *N. bostrycoides* grown in PDA after 14 days at 25 °C (A); macroconidium (B); microconidia (C); Front and back from culture of *N. aracacearum* grown in PDA after 12 days at 25 °C (D); acervulus (E); conidia (F); Front from culture of *N. aracacearum* grown in PDA after 12 days at 25 °C under near light (G); pycnidia (H); conidia (I); horizontal section of the stroma showing the conidioma (J); conidiogenic cells (K).

been reported to cause canker in Japanese pear (*Pyrus pyrifolia*) and as an endophyte in brazilwood (*Caesalpinia echinata*) (Becker, 2003; Cota et al., 2018). In this study, we observed *N. pseudotrachia* causing branch dieback in avocados for the first time.

Cytospora is a genus that causes diseases in the branches and trunks of tree species, with symptoms appearing as sunken, discolored areas on the bark of the stem and subsequent plant death (Fan et al., 2020; Lawrence et al., 2018; Pan et al., 2021; Rumbos, 1988). *Cytospora viridistroma* causes disease in eucalyptus (Adams et al., 2004; Jiang et al., 2020), but this is the first report of it causing dieback in avocado worldwide.

All isolates showed symptoms on the stems of seedlings, indicating colonization of vascular tissues, a common characteristic of the genera studied in this work. These genera are widely known to cause canker and dieback (Jiang et al., 2018; Lawrence et al., 2018; Pan et al., 2020; Yang et al., 2018; Crespo et al., 2019). In fruits inoculated with *Cytospora* and *Neopestalotiopsis*, lesions remain restricted to the inoculation site. Although lesions on fruits caused by these fungi are uncommon, some

reports confirm these findings (Ayoubi and Soleimani, 2015; Baggio et al., 2021; Mincuzzi et al., 2017; Venter et al., 2017).

Dieback is a complex symptom that can be associated with various causal agents, exhibiting similar symptomatology (Arjona-Girona et al., 2019; Hilário et al., 2020; Ismail et al., 2012; McDonald et al., 2009; Rodríguez-Gálvez et al., 2017). This makes the diagnosis and effective control of the disease particularly challenging. Accurate identification of the causal agents of avocado dieback in Brazil is essential for monitoring and providing recommendations for effective control methods for avocado growers (Fischer and Firmino, 2023).

The management of dieback is based on practices that prioritize protection and exclusion, aiming to delay damage caused by disease progression (Saeed et al., 2017; Tavares et al., 2005). Generally, it is recommended to remove and dispose of infected plant parts, eliminate crop residues, perform pruning, disinfect tools, and control damaged insects. Once established, the disease can be reduced through applications of triazole, strobilurin, phenylpyridinyloamine, and phenylpyrrole fungicides, as well as the biological control agent *Bacillus velezensis*

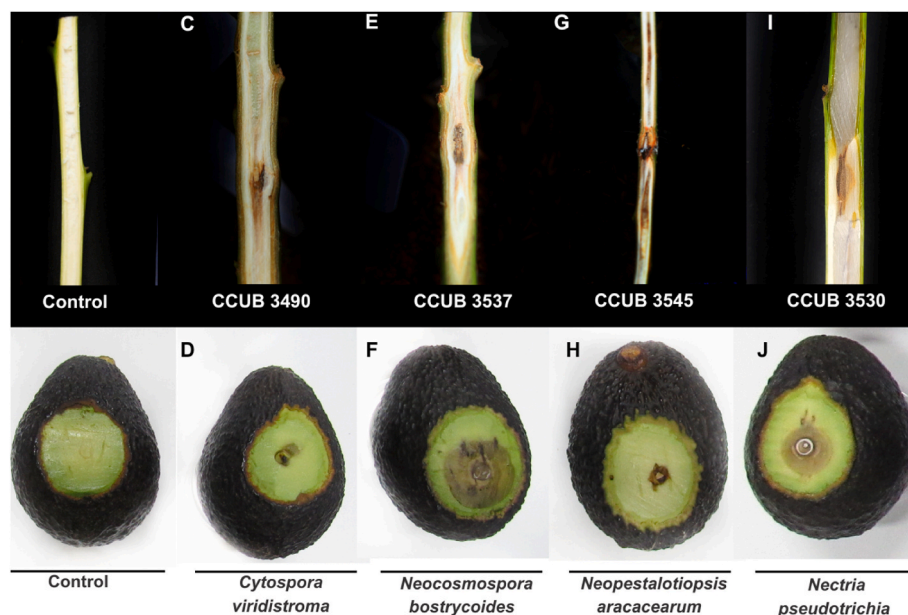


Fig. 6. Stems of the “Margarida” variety and fruits of the “Hass” avocado inoculated with isolates of *Cytospora* (C and D), *Neocosmospora* (E and F), *Neopestalotiopsis* (G and H), *Nectria* (I and J) and control (A and B). Stems were inoculated using the mycelium disc method and evaluated after 40 days. Fruits were inoculated with mycelium disc directly into the pulp, with evaluation after 7 days.

(Bernal et al., 2025; Li et al., 2024; Tavares et al., 2005; Twizeyimana et al., 2013).

Identifying emerging pathogens that may cause losses is crucial for avocado-producing countries with different climatic conditions, as it helps understand the geographic distribution of these phytopathogens and the conditions that favor their development.

5. Conclusion

To our knowledge, this is the first report of *Neopestalotiopsis arecaeum*, *Neocosmospora bostrycoides*, *Nectria pseudotrachia* and *Cytospora viridistroma* causing avocado dieback in Brazil and worldwide. Since dieback is exacerbated by nutritional deficiencies, drought stress, elevated temperatures, frost, and over-irrigation (Valencia et al., 2022), these conditions should be avoided or mitigated. This should be combined with branch protection using fungicides and proper sanitation of pruning tools to prevent the proliferation of these fungal species.

CRedit authorship contribution statement

Thais França Silva: Writing – original draft, Visualization, Investigation, Data curation, Conceptualization. **João Lucas Pimentel Duarte:** Writing – review & editing, Visualization, Investigation. **Jefferson Bertin Vélez-Olmedo:** Writing – review & editing, Visualization, Investigation. **Willie Anderson dos Santos Vieira:** Writing – review & editing, Visualization, Software. **Luiz Eduardo Bassay Blum:** Writing – review & editing, Visualization, Investigation, Conceptualization. **Danilo Batista Pinho:** Writing – review & editing, Visualization, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors of the manuscript are Thais França Silva, João Lucas Pimentel Duarte, Jefferson Bertin Vélez Olmedo; Willie Anderson dos Santos Vieira; Luiz Eduardo Bassay Blum, Danilo Batista Pinho.

We declare that this manuscript has not been published in another journal and that all of the authors agree to submit it to Crop Protection. If we can provide you with any additional information, please do not hesitate to contact us.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cropro.2025.107168>.

Data availability

Data will be made available on request.

References

- Adams, G., Wingfield, M.J., Common, R., Roux, J., 2004. Phylogenetic Relationships and Morphology of *Cytospora* Species and Related Teleomorphs (Ascomycota, Diaporthales, Yalsaceae) from *Eucalyptus*-Preface.
- Arjona-Girona, I., Ruano-Rosa, D., López-Herrera, C.J., 2019. Identification, pathogenicity and distribution of the causal agents of dieback in avocado orchards in Spain. Span. j. agric. res. 17. <https://doi.org/10.5424/sjar/2019171-13561>.
- Auger, J., Palma, F., Pérez, I., Esterio, M., 2013. First Report of *Neofusicoccum australe* (*Botryosphaeria australis*), as a branch dieback pathogen of avocado trees in Chile. Plant Dis. 97, 842. <https://doi.org/10.1094/PDIS-12-12-0980-PDN>.
- Ayoubi, N., Soleimani, M.J., 2015. Strawberry fruit rot caused by *Neopestalotiopsis iranensis* sp. nov., and *N. mesopotamica*. Curr. Microbiol. <https://doi.org/10.1007/s00284-015-0955-y>.
- Baggio, J.S., Forcelini, B.B., Wang, N.-Y., Ruschel, R.G., Mertely, J.C., Peres, N.A., 2021. Outbreak of leaf spot and fruit rot in Florida strawberry caused by *Neopestalotiopsis* spp. Plant Dis. 105, 305–315. <https://doi.org/10.1094/PDIS-06-20-1290-RE>.
- Becker, W.F., 2003. *Nectria pseudotrachia*, como agente causal de cancro de ramos, ocorrendo em pereira japonesa no Brasil. Fitopatol. Bras. 28, 107. <https://doi.org/10.1590/S0100-41582003000100018>.
- Bernal, V.V., Peghrad, Z., Kamali Dashtarzhaneh, M., Khodadadi, F., 2025. Identification, characterization, and fungicide sensitivity of *Botryosphaeriaceae* fungi associated with avocado branch canker disease in southern California. Plant Dis. <https://doi.org/10.1094/PDIS-12-24-2674-RE>.
- Çalış, Ö., Çelik, S., Fidan, H., Tek, M.I., Shah, M., Tozlu, I., Wani, S.H., 2024. Emerging pathogens and disease dynamics threatening avocado production in southern

- Türkiye. J. Plant Dis. Prot. 131, 1653–1663. <https://doi.org/10.1007/s41348-024-00954-6>.
- Camarero, J.J., Gazol, A., Sangüesa-Barreda, G., Oliva, J., Vicente-Serrano, S.M., 2015. To die or not to die: Early warnings of tree dieback in response to a severe drought. J. Ecol. 103, 44–57. <https://doi.org/10.1111/1365-2745.12295>.
- Carbone, I., Kohn, L.M., 1999. A method for designing primer sets for speciation studies in filamentous ascomycetes. Mycologia 91, 553–556. <https://doi.org/10.1080/00275514.1999.12061051>.
- Cota, B.B., Tunes, L.G., Maia, D.N.B., Ramos, J.P., Oliveira, D.M. de, Kohlhoff, M., Alves, T.M. de A., Souza-Fagundes, E.M., Campos, F.F., Zani, C.L., 2018. Leishmanicidal compounds of *Nectria pseudotrichia*, an endophytic fungus isolated from the plant *Caesalpinia echinata* (Brazilwood). Mem. Inst. Oswaldo Cruz 113, 102–110. <https://doi.org/10.1590/0074-02760170217>.
- Crespo, M., Lawrence, D., Nouri, M., Doll, D., Trouillas, F., 2019. Characterization of *Fusarium* and *Neocosmospora* species associated with crown rot and stem canker of pistachio rootstocks in California. Plant Dis 103, 1931–1939. <https://doi.org/10.1094/PDIS-11-18-2012-RE>.
- De Hoog, G.S., Gerrits Van Den Ende, A.H.G., 1998. Molecular diagnostics of clinical strains of filamentous Basidiomycetes. Mycoses 41, 183–189. <https://doi.org/10.1111/j.1439-0507.1998.tb00321.x>.
- Fan, X.L., Bezerra, J.D.P., Tian, C.M., Crous, P.W., 2020. *Cytospora* (diaporthales) in China. Persoonia: Molecular Phylogeny and Evolution of Fungi 45, 1–45. <https://doi.org/10.3767/persoonia.2020.45.01>.
- FAO, 2021. OECD-FAO Agricultural Outlook 2021–2030. OECD. <https://doi.org/10.1787/19428846-en>.
- Fiorenza, A., Gusella, G., Aiello, D., Polizzi, G., Voglmayr, H., 2022. *Neopestalotiopsis siciliana* sp. nov. and *N. rosae* Causing stem lesion and dieback on avocado plants in Italy. J. Fungi 8. <https://doi.org/10.3390/jof8060562>.
- Fiorenza, A., Gusella, G., Vecchio, L., Aiello, D., Polizzi, G., 2023. Diversity of Botryosphaeriaceae species associated with canker and dieback of avocado (*Persea americana*) in Italy. Phytopathol. Mediterr. 62, 47–63. <https://doi.org/10.36253/phyto-14057>.
- Fischer, I.H., Firmino, A.C., 2023. Main diseases of avocado in Brazil. RAPP 29, 106–130. <https://doi.org/10.31976/0104-038321v290005>.
- Flores, M., Saravia, C., Vergara, C.E., Avila, F., Valdés, H., Ortiz-Viedma, J., 2019. Avocado oil: characteristics, properties, and applications. Molecules. <https://doi.org/10.3390/molecules24112172>.
- Glass, N.L., Donaldson, G.C., 1995. Development of primer sets designed for use with the PCR to amplify conserved genes from filamentous ascomycetes. Appl. Environ. Microbiol. 61, 1323–1330. <https://doi.org/10.1128/aem.61.4.1323-1330.1995>.
- Guarnaccia, V., Aiello, D., Papadantonakis, N., Polizzi, G., Gullino, M.L., 2022. First report of branch cankers on avocado (*Persea americana*) caused by *Neocosmospora* (syn. *Fusarium*) *persea* in Crete (Greece). J. Plant Pathol. 104, 419–420. <https://doi.org/10.1007/s42161-021-00983-2>.
- Guarnaccia, V., Sandoval-Denis, M., Aiello, D., Polizzi, G., Crous, P.W., 2018. *Neocosmospora persea* sp. nov., causing trunk cankers on avocado in Italy. Fungal Syst. Evol. 1, 131–140. <https://doi.org/10.3114/fuse.2018.01.06>.
- Gusella, G., Gugliuzzo, A., Guarnaccia, V., Martino, I., Aiello, D., Costanzo, M.B., Russo, A., Groenewald, J.Z., Crous, P.W., Polizzi, G., 2024. Fungal species causing canker and wilt of *Ficus carica* and evidence of their association by bark beetles in Italy. Plant Dis. 108, 2136–2147. <https://doi.org/10.1094/PDIS-01-24-0251-RE>.
- Guterres, D.C., Silva, M.A., Martins, M.D., Azevedo, D.M.Q., Lisboa, D.O., Pinho, D.B., Furtado, G.Q., 2023. Leaf spot caused by *Neopestalotiopsis* species on Arecaceae in Brazil. Australas. Plant Pathol. 52, 47–62. <https://doi.org/10.1007/s13313-022-00893-6>.
- Hernández, D., García-Pérez, O., Perera, S., González-Carracedo, M.A., Rodríguez-Pérez, A., Siverio, F., 2023. Fungal pathogens associated with aerial symptoms of avocado (*Persea americana* Mill.) in tenerife (canary islands, Spain) focused on species of the family botryosphaeriaceae. Microorganisms 11, 585. <https://doi.org/10.3390/microorganisms11030585>.
- Hilário, S., Lopes, A., Santos, L., Alves, A., 2020. Botryosphaeriaceae species associated with blueberry stem blight and dieback in the Centre Region of Portugal. Eur. J. Plant Pathol. 156, 31–44. <https://doi.org/10.1007/s10658-019-01860-6>.
- Hrycan, J., Hart, M., Bowen, P., Forge, T., Úrbez-Torres, J.R., 2020. Grapevine trunk disease fungi: their roles as latent pathogens and stress factors that favour disease development and symptom expression. Phytopathol. Mediterr. 59, 395–424. <https://doi.org/10.14601/Phyto-11275>.
- Hultberg, T., Sandström, J., Felton, A., Öhman, K., Rönnberg, J., Witzell, J., Cleary, M., 2020. Ash dieback risks an extinction cascade. Biol. Conserv. 244. <https://doi.org/10.1016/j.biocon.2020.108516>.
- Instituto Brasileiro de Geografia e Estatística, 2024. Produção de Abacate [Produção de Abacate no Brasil | IBGE]. IBGE.
- Ismail, A.M., Cirvilleri, G., Polizzi, G., Crous, P.W., Groenewald, J.Z., Lombard, L., 2012. *Lasiodiplodia* species associated with dieback disease of mango (*Mangifera indica*) in Egypt. Australas. Plant Pathol. 41, 649–660. <https://doi.org/10.1007/s13313-012-0163-1>.
- Jacobs, K., Bergdahl, D.R., Wingfield, M.J., Halik, S., Seifert, K.A., Bright, D.E., Wingfield, B.D., 2004. *Leptographium wingfieldii* introduced into North America and found associated with exotic *Tomicus piniperda* and native bark beetles. Mycol. Res. 108, 411–418. <https://doi.org/10.1017/S09593756204009748>.
- Jiang, N., Bonthond, G., Fan, X.L., Tian, C.M., 2018. *Neopestalotiopsis rosicola* sp. nov. causing stem canker of *Rosa chinensis* in China. Mycotaxon 133, 271–283. <https://doi.org/10.5248/133.271>.
- Jiang, N., Fan, X., Tian, C., Crous, P.W., 2020. Reevaluating cryphonectriaceae and allied families in diaporthales. Mycologia 112, 267–292. <https://doi.org/10.1080/00275514.2019.1698925>.
- Jurskis, V., 2005. Eucalypt decline in Australia, and a general concept of tree decline and dieback. Ecol. Manag. 215, 1–20. <https://doi.org/10.1016/j.foreco.2005.04.026>.
- Katoh, K., Standley, D.M., 2013. MAFFT multiple sequence alignment software version 7: improvements in performance and usability. Mol. Biol. Evol. 30, 772–780. <https://doi.org/10.1093/molbev/mst010>.
- Lawrence, D.P., Holland, L.A., Nouri, M.T., Travadon, R., Abramians, A., Michailides, T. J., Trouillas, F.P., 2018. Molecular phylogeny of *Cytospora* species associated with canker diseases of fruit and nut crops in California, with the descriptions of ten new species and one new combination. IMA Fungus 9, 333–370. <https://doi.org/10.5598/imafungus.2018.09.02.07>.
- Li, X., Tan, C., Li, P., Lin, L., Zhou, J., Tao, H., Cai, Y., 2024. Biological control of avocado branch blight caused by *Lasiodiplodia theobromae* using *Bacillus velezensis*. Plant Dis. 108, 2053–2064. <https://doi.org/10.1094/PDIS-10-23-2216-RE>.
- Lin, C.-H., Dong, P.-P., Fang, S.-Q., Li, M.-F., Liu, W.-B., Miao, W.-G., 2018. First report of avocado dieback disease caused by *Pestalotiopsis longiseta* in China. Plant Dis. 102, 2660. <https://doi.org/10.1094/PDIS-06-18-0939-PDN>.
- Liu, Y.J., Whelen, S., Hall, B.D., 1999. Phylogenetic relationships among ascomycetes: evidence from an RNA Polymerase II Subunit. Mol. Biol. Evol. 16, 1799–1808.
- McDonald, V., Lynch, S., Eskalen, A., 2009. First report of *Neofusicoccum australe*, *N. luteum*, and *N. parvum* associated with avocado branch canker in California. Plant Dis. 93, 967. <https://doi.org/10.1094/PDIS-93-9-0967B>.
- Miller, M.A., Pfeiffer, W., Schwartz, T., 2010. Creating the CIPRES Science Gateway for inference of large phylogenetic trees. In: 2010 Gateway Computing Environments Workshop. GCE. <https://doi.org/10.1109/GCE.2010.5676129>, 2010.
- Mincuzzi, A., Sanzani, S.M., Garganese, F., Ligorio, A., Ippolito, A., 2017. First report of *Cytospora punicea* causing fruit rot on pomegranate in Italy. J. Plant Pathol. 99, 302.
- Ninos, T.F., Veloso, J.S., da Silva, M.A., da Paz, C.D., Câmara, M.P.S., Peixoto, A.R., 2021. Occurrence of *Fusarium bostrychoides* as cause of wilt on yellow passion fruit plants in Brazil. J. Plant Pathol. <https://doi.org/10.1007/s42161-021-00928-9>.
- O'donnell, K., Cigelnik, E., 1997. Two Divergent Intragenomic rDNA ITS2 Types within a Monophyletic Lineage of the Fungus *Fusarium* Are Nonorthologous.
- Oliveira, V.C., Silva, H.A.O., Mendes, F.Q., Oliveira, E.B., Abreu, L.M., Vieira, E.N.R., 2024. First report of *Colletotrichum nymphaeae* causing anthracnose on Hass avocados in Brazil. Plant Dis. <https://doi.org/10.1094/PDIS-08-24-1674-PDN>.
- Pan, M., Zhu, H., Bonthond, G., Tian, C., Fan, X., 2020. High diversity of *Cytospora* associated with canker and dieback of rosaceae in China, with 10 new species described. Front. Plant Sci. 11. <https://doi.org/10.3389/fpls.2020.00690>.
- Pan, M., Zhu, H., Liang, L., Tian, C., Fan, X., 2021. Studies of canker and dieback of oak tree in China, with two *Cytospora* species described. Plant Pathol. 70, 2005–2015. <https://doi.org/10.1111/ppa.13435>.
- Piccinin, E., Pascholati, S.F., 2018. Doenças do abacateiro. In: Amorim, L., Rezende, J.A.M., Bergamino Filho, A. (Eds.), Manual de Fitopatologia. Editora Agronômica Ceres Ltda, Ouro Fino.
- Pinho, D.B., Firmino, A.L., Ferreira-Junior, W.G., Pereira, O.L., 2013. An efficient protocol for DNA extraction from *Meliolales* and the description of *Meliola centellae* sp. nov. Mycotaxon 122, 333–345. <https://doi.org/10.5248/122.333>.
- Ploetz, R.C., Hughes, M.A., Kendra, P.E., Fraedrich, S.W., Carrillo, D., Stelinski, L.L., Hulcr, J., Mayfield, A.E., Dreaden, T.J., Crane, J.H., Evans, E.A., Schaffer, B.A., Rollins, J.A., 2017. Recovery plan for laurel wilt of avocado, caused by *Raffaelea lauricola*. Plant Health Prog. 18, 51–77. <https://doi.org/10.1094/PHP-12-16-0070-RP>.
- Ramírez-Gil, J.G., Peterson, A.T., 2019. Current and potential distributions of the most important diseases affecting Hass avocado in Antioquia Colombia. J. Plant Protect. Res. 59. <https://doi.org/10.24425/jppr.2019.129288>.
- Rodríguez-Gálvez, E., Guerrero, P., Barradas, C., Crous, P.W., Alves, A., 2017. Phylogeny and pathogenicity of *Lasiodiplodia* species associated with dieback of mango in Peru. Fungal Biol. 121, 452–465. <https://doi.org/10.1016/j.funbio.2016.06.004>.
- Ronquist, F., Huelsenbeck, J.P., 2003. MrBayes 3: Bayesian phylogenetic inference under mixed models. Bioinformatics 19, 1572–1574. <https://doi.org/10.1093/bioinformatics/btg180>.
- Rumbos, I.C., 1988. *Cytospora oleina* causing canker and dieback of olive in Greece. Plant Pathol. 37, 441–444. <https://doi.org/10.1111/j.1365-3059.1988.tb02098.x>.
- Saeed, E.E., Sham, A., Abuzarqa, A., Shurafa, K.A.A., Naqbi, T.S.A., Itrat, R., El-Tarabily, K., Abuqamar, S.F., 2017. Detection and management of mango dieback disease in the United Arab Emirates. Int. J. Mol. Sci. 18, 2086. <https://doi.org/10.3390/ijms18102086>.
- Serrato-Díaz, L.M., Mariño, Y.A., González, J. de J., Goenaga, R., Bayman, P., 2024. Coffee fruit rot: the previously unrecognized role of *Fusarium* and its interactions with the coffee berry borer (*Hypothenemus hampei*). Phytopathology 114, 1320–1332. <https://doi.org/10.1094/PHYTO-02-24-0046-R>.
- Soares, M.G.O., Alves, E., Silveira, A.L., Pereira, F.D., Guimarães, S.C., 2021. *Colletotrichum siamense* is the main aetiological agent of anthracnose of avocado in south-eastern Brazil. Plant Pathol. 70, 154–166. <https://doi.org/10.1111/ppa.13262>.
- Sommaruga, R., Eldridge, H.M., 2021. Avocado production: water footprint and socio-economic implications. EuroChoices 20, 48–53. <https://doi.org/10.1111/1746-692X.12289>.
- Stamatakis, A., 2014. RAxML version 8: a tool for phylogenetic analysis and post-analysis of large phylogenies. Bioinformatics 30, 1312–1313. <https://doi.org/10.1093/bioinformatics/btu033>.
- Sung, G.H., Sung, J.M., Hywel-Jones, N.L., Spatafora, J.W., 2007. A multi-gene phylogeny of Clavicipitaceae (Ascomycota, Fungi): identification of localized incongruence using a combinational bootstrap approach. Mol. Phylogenet. Evol. 44, 1204–1223. <https://doi.org/10.1016/j.ympev.2007.03.011>.
- Tavares, S.C.C.H., Costa, V.S.O., Moreira, A.N., 2005. Manejo de morte descendente. Instruções técnicas da Embrapa Semi-Árido. Petrolina, p. 2.

- Trakunyingcharoen, T., Cheewangkoon, R., To-anun, C., 2015. Phylogenetic study of the Botryosphaeriaceae species associated with avocado and par rubber in Thailand. *Chiang Mai J. Sci.* 42, 104–116.
- Twizeyimana, M., McDonald, V., Mayorquin, J.S., Wang, D.H., Na, F., Akgül, D.S., Eskalen, A., 2013. Effect of fungicide application on the management of avocado branch canker (formerly Dothiorella canker) in California. *Plant Dis.* 97, 897–902. <https://doi.org/10.1094/PDIS-06-12-0518-RE>.
- Vaidya, G., Lohman, D.J., Meier, R., 2011. SequenceMatrix: concatenation software for the fast assembly of multi-gene datasets with character set and codon information. *Cladistics* 27, 171–180. <https://doi.org/10.1111/j.1096-0031.2010.00329.x>.
- Valencia, A.L., Gil, P.M., Latorre, B.A., Rosales, I.M., 2019. Characterization and pathogenicity of Botryosphaeriaceae species obtained from avocado trees with branch canker and dieback and from avocado fruit with stem end rot in Chile. *Plant Dis.* 103, 996–1005. <https://doi.org/10.1094/PDIS-07-18-1131-RE>.
- Valencia, A.L., Saavedra-Torrico, J., Rosales, I.M., Mártiz, J., Retamales, A., Link, A., Gil, P.M., 2022. Unveiling the predisposing factors for the development of branch canker and dieback in avocado: a case of study in Chilean orchards. *Horticulturae* 8, 1121. <https://doi.org/10.3390/horticulturae8121121>.
- Valor, T., Camprodon, J., Buscarini, S., Casals, P., 2020. Drought-induced dieback of riparian black alder as revealed by tree rings and oxygen isotopes. *Ecol. Manag.* 478. <https://doi.org/10.1016/j.foreco.2020.118500>.
- van den Berg, N., du Toit, M., Morgan, S.W., Fourie, G., de Beer, Z.W., 2019. First report of *Fusarium euwallaceae* causing necrotic lesions on *Persea americana* in South Africa. *Plant Dis.* 103, 1774. <https://doi.org/10.1094/PDIS-10-18-1818-PDN>.
- Venter, E., Lennox, C.L., Meitz-Hopkins, J.C., 2017. First report of *Cytospora punicae* causing post-harvest fruit rot on pomegranate in South Africa. *Plant Dis.* 101, 631. <https://doi.org/10.1094/PDIS-08-16-1157-PDN>.
- Vilgalys, R., Hester, M., 1990. Rapid genetic identification and mapping of enzymatically amplified ribosomal DNA from several *Cryptococcus* species. *J. Bacteriol.* 172, 4238–4246. <https://doi.org/10.1128/jb.172.8.4238-4246.1990>.
- Yang, Q., Du, Z., Liang, Y.M., Tian, C.M., 2018. Molecular phylogeny of *Nectria* species associated with dieback and canker diseases in China, with a new species described. *Phytotaxa* 356, 199–214. <https://doi.org/10.11646/phytotaxa.356.3.2>.